DAY 04 · THEORY HANDOUT

Matplotlib: Build Clear Charts

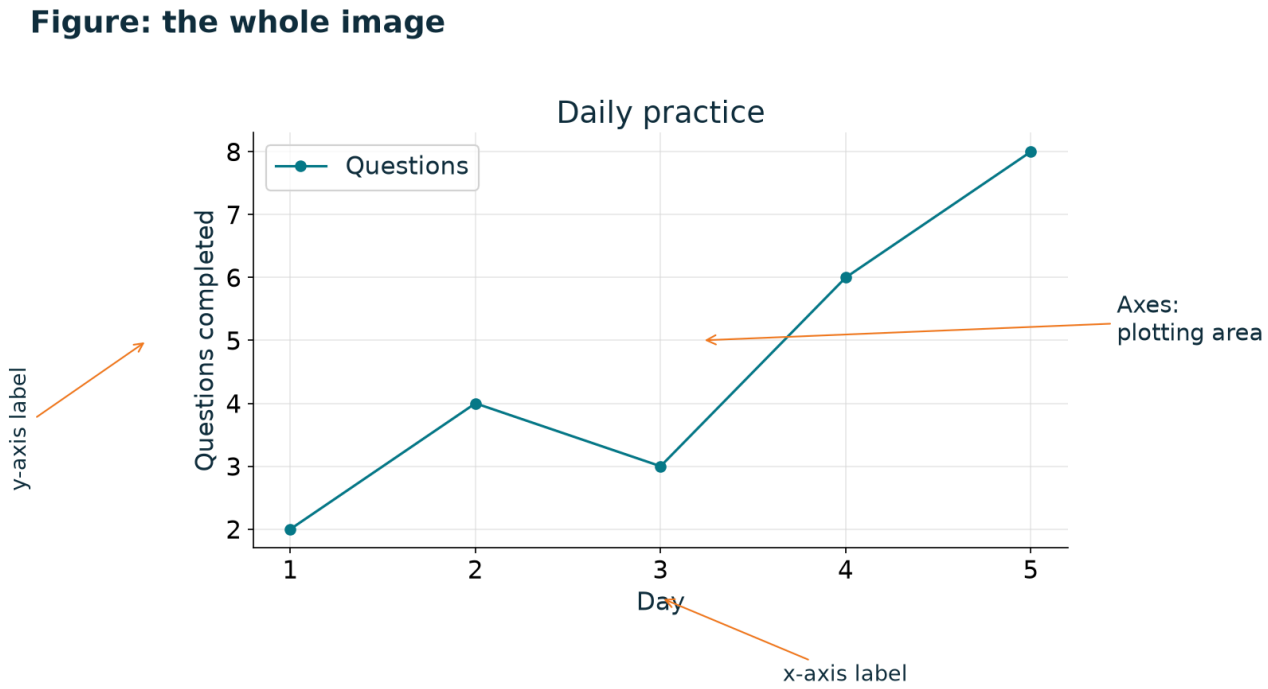
We read actual charts, name their parts, and connect each plotting choice to the data.

KGP Talkie

Scan to subscribe

Laxmi Kant Tiwari · kgptalkie.com

Figure, Axes, and Axis



The Figure holds the whole image. One Axes holds one plotting area.

Definition

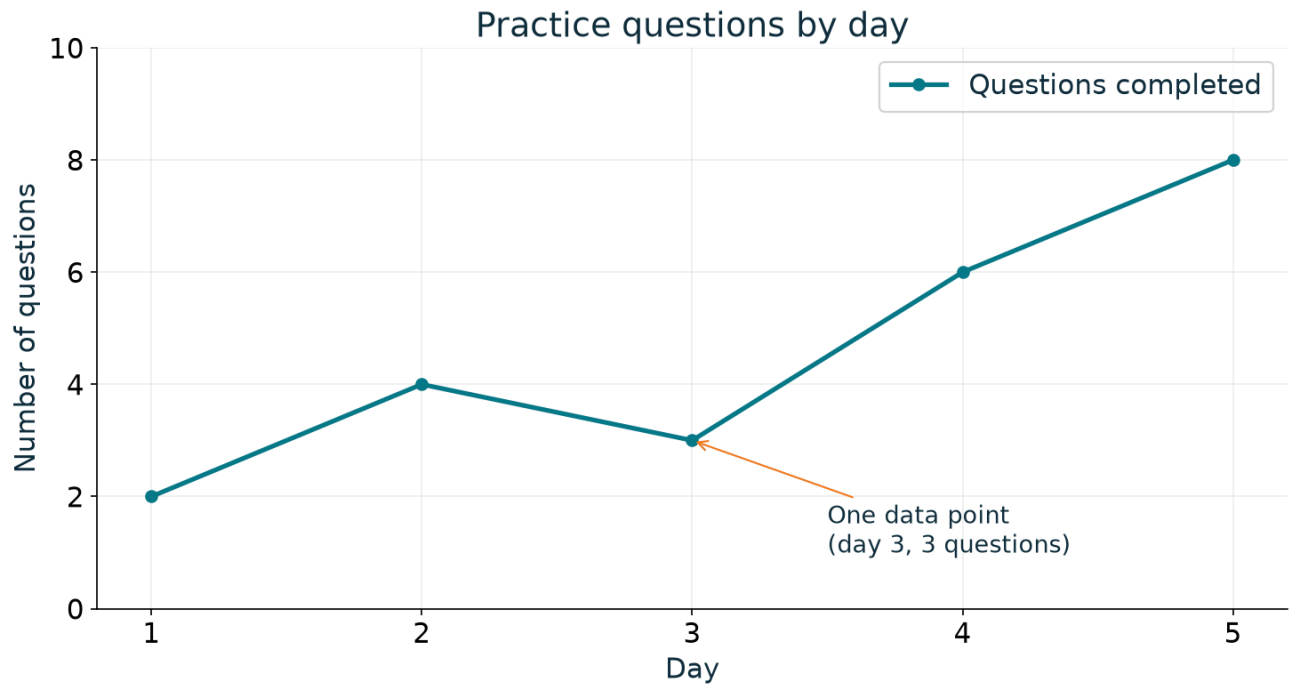
Matplotlib is a Python library for data visualisation. A plot maps values to marks such as lines, bars, or points.

How it works

Axes is Matplotlib's name for one plotting area. An **axis** is a scale along one direction. Tick labels show values on that scale.

> The names Axes and axis describe different parts. Find both in the diagram.

Line plots



The point at day 3 has a value of three questions. The line joins days in order.

Definition

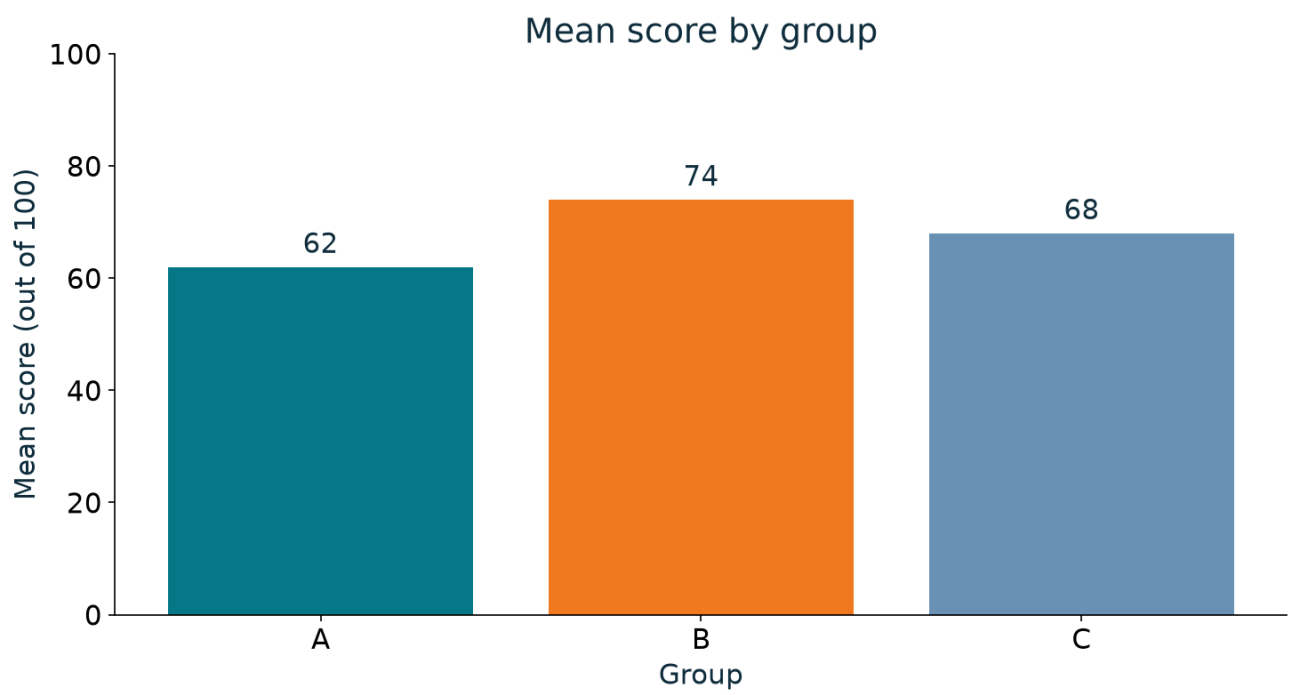
A **line plot** displays change across an ordered sequence, often time. Each point pairs an x value with a y value.

How it works

plt.plot(days, questions) draws the line. A circle marker shows the measured values. The connecting line makes the order visible.

> The line falls from day 2 to day 3, then rises. Read the values from the axes.

Bar plots



Each category gets one bar. The bar height represents its mean score.

Definition

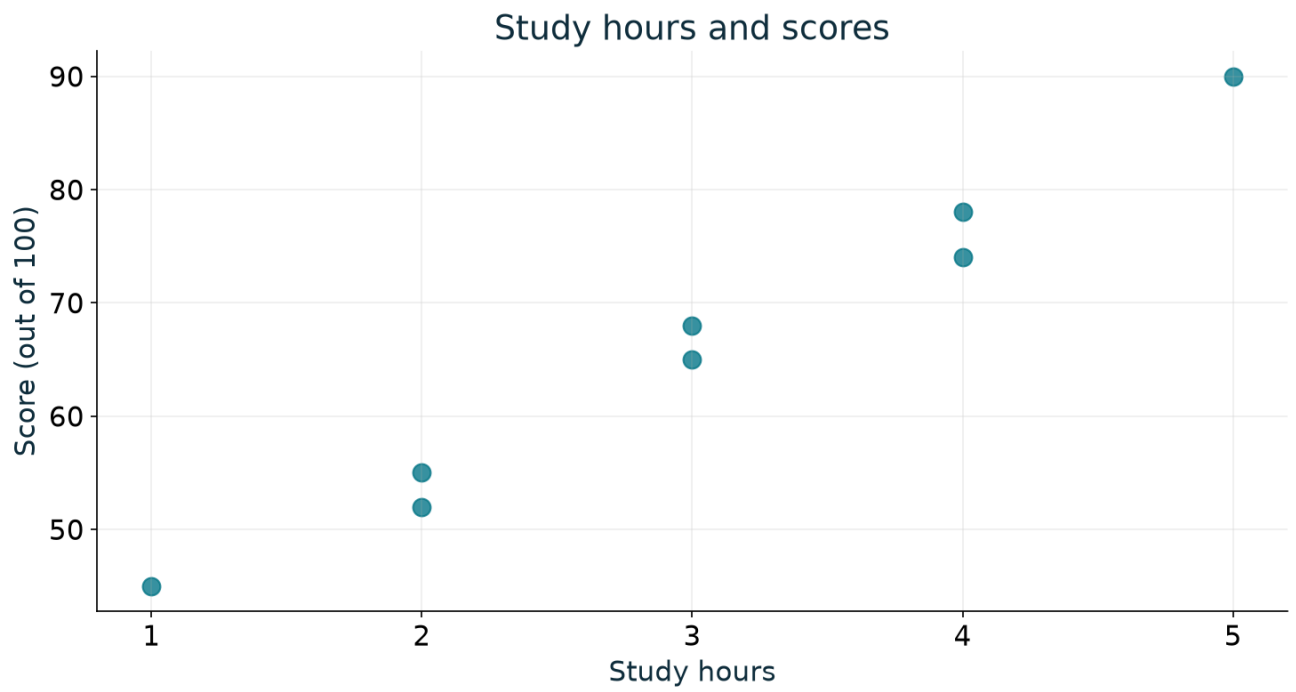
A **bar plot** compares separate categories. The value axis normally begins at zero because the bar length carries the comparison.

How it works

Group B has mean 74. Group A has mean 62. The difference is 12 score points. The plotted means do not show the spread within each group.

Use a bar plot for categories such as groups or products. Use a line when the sequence matters.

Scatter plots



One point is one student. The two coordinates give study hours and score.

Definition

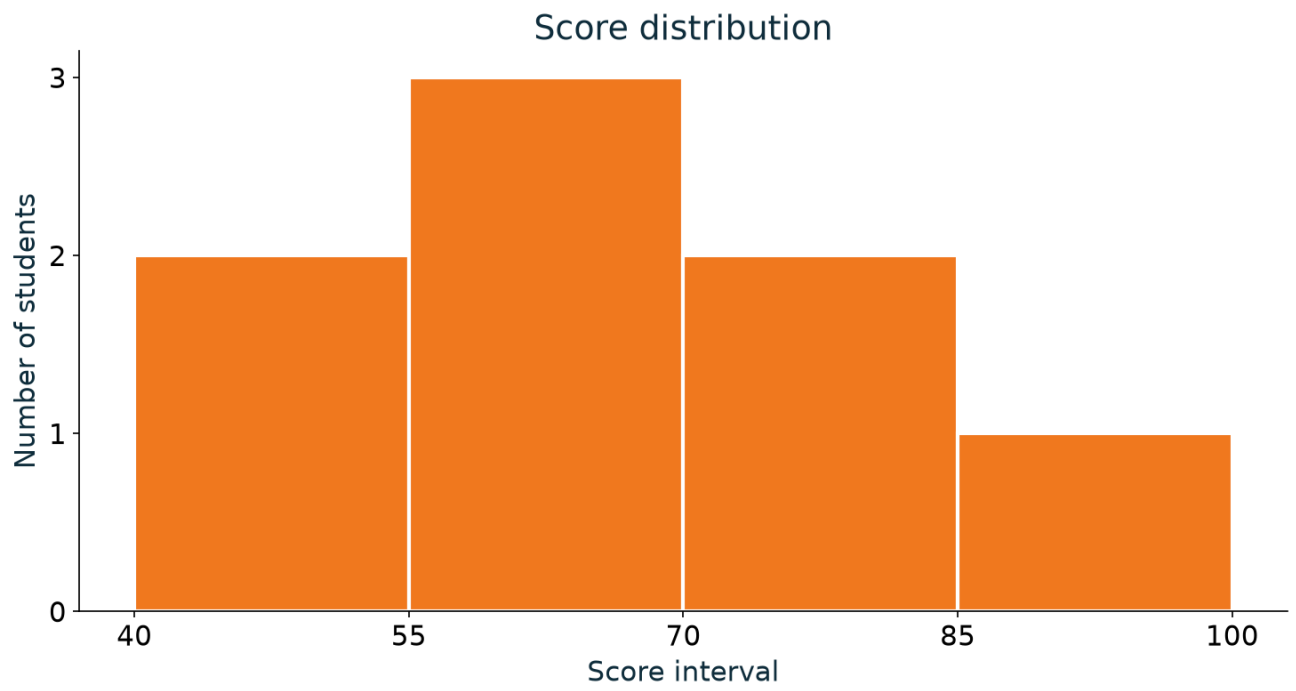
A **scatter plot** displays the relationship between two numeric variables. It can show clusters, unusual points, and broad patterns.

How it works

The example points tend to rise with hours. This is an association in a small invented dataset. It does not prove that hours alone caused the scores.

> Read the point at (5, 90): five study hours and a score of 90.

Histograms and bins



Bars count values in numeric intervals. Here the first bin contains scores 45 and 52.

Definition

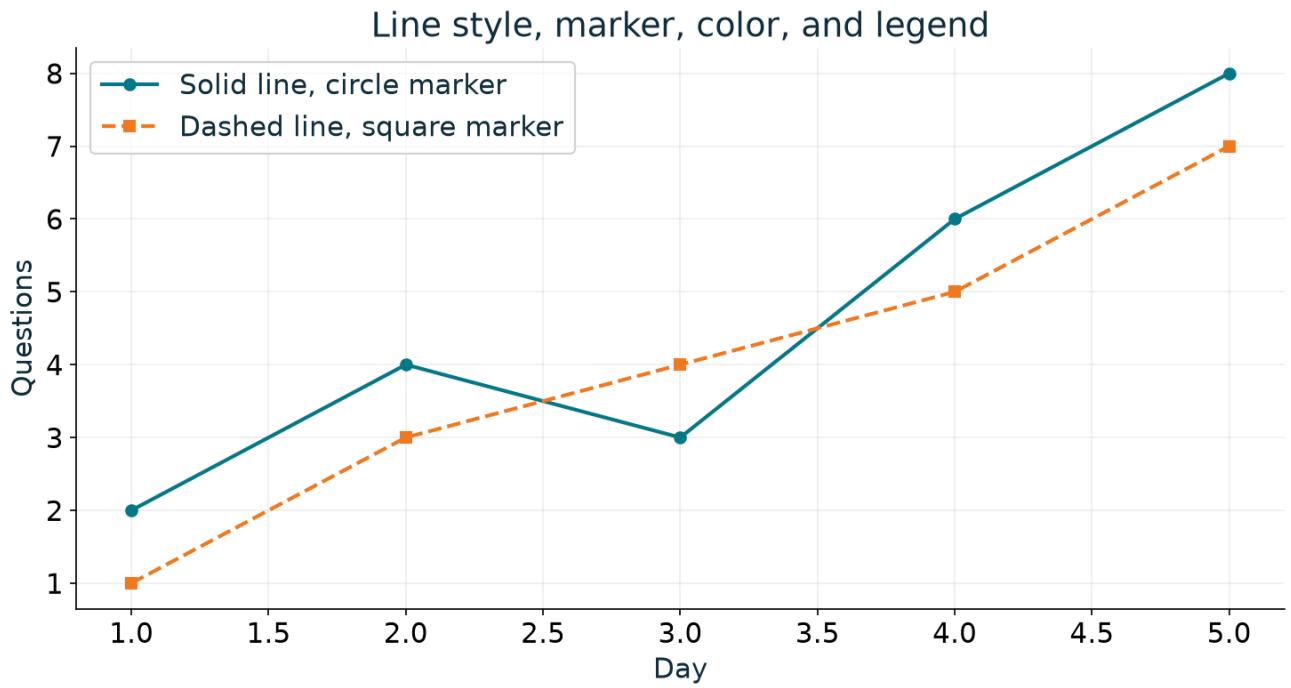
A **histogram** shows a distribution. A **bin** is a numeric interval. Bar height is the count of observations in that interval.

How it works

Our bin edges are 40, 55, 70, 85, and 100. Changing edges can change the look of a histogram, so inspect the data too.

> The interval from 55 up to 70 contains 55, 65, and 68: three students.

Style and legends



Color, line style, and marker shape distinguish the two series.

Definition

A **legend** identifies named data series. A **marker** shows an individual data point. Line style changes the connecting line.

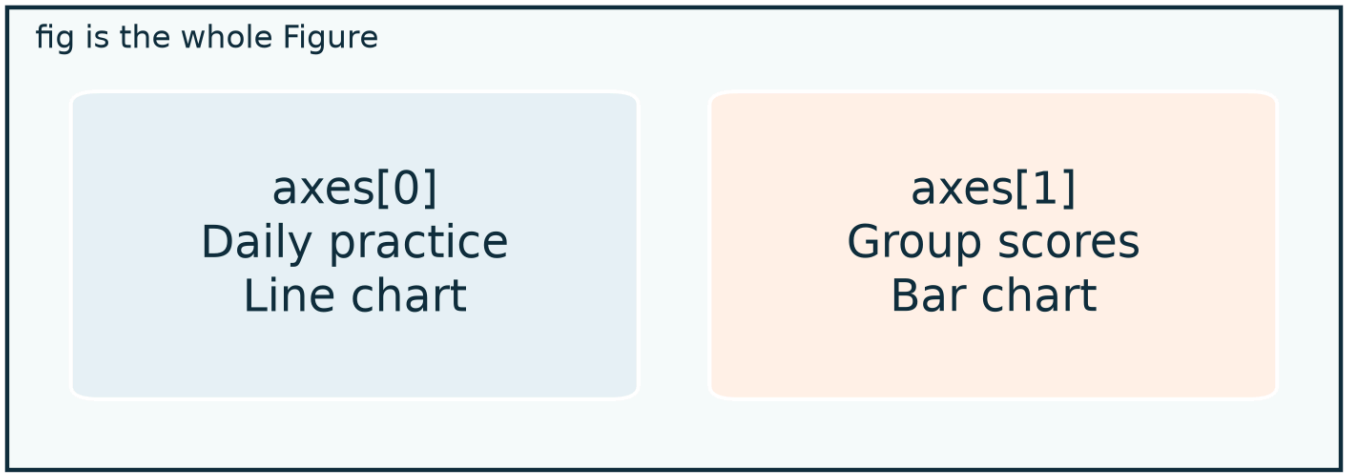
How it works

color chooses a color. marker="o" gives circles. linestyle="--" gives dashes. linewidth controls thickness. label provides the legend text.

> Use visual differences with a purpose. Keep group colors consistent across related charts.

Subplots and layout

subplots(1, 2): one Figure, two separate Axes



Use axes[0].plot(...) for the left chart and axes[1].bar(...) for the right.

The Figure is the whole canvas. Each Axes holds one chart and its own title and axis labels.

Definition

Subplots arrange multiple plotting areas in one figure. plt.subplots(1, 2) makes one row and two columns.

How it works

subplots(1, 2) returns a Figure and two Axes. axes[0] holds the left chart. axes[1] holds the right chart. Titles and labels belong to each Axes.

> The two y axes measure different things. Read each label before comparing the plots.

Labels, limits, and saving

Tool	Purpose	Example
set / xlabel / ylabel	Name variables and units	Study hours; score out of 100
xlim / ylim	Set visible axis limits	ylim(0, 100)
grid	Add guide lines	grid(alpha=0.25)
figsize	Choose image size in inches	figsize=(7, 4)
savefig	Save an image file	fig.savefig("plot.png", dpi=150)

DPI means dots per inch. More DPI gives more pixels at the same figure size.

Definition

A readable plot has a clear subject, labeled units, and a scale that supports the comparison.

How it works

Set labels before saving. Keep a Figure reference so you can save the intended chart. A chart can be misleading when important values are hidden by the chosen limits.

> For score bars, a zero baseline and a 0 to 100 range make the scale clear.

DAY 04 · COMPLETE

Thank You

Follow KGP Talkie for new lessons, course updates and live sessions.

Scan to follow

Instagram
@kgptalkie

WhatsApp Channel
KGP Talkie